

BUILGATES Inputs

3. Data Lakehouse Architecture

GATES P2 implements a three-tier medallion architecture on MinIO object storage running on COARE. Apache Iceberg is used as the open table format for silver and gold layers, providing time-travel, schema evolution, and partition pruning capabilities.

Data Partitioning and Retention

Bronze files are partitioned by source agency, dataset name, and ingestion date. Retention at the bronze tier is indefinite to support full auditability. Silver and gold Iceberg tables are partitioned by domain and time granularity appropriate to each use case. Lineage is registered at the silver tier, linking each conformed table to its bronze source partition.

API Exposure

Only the gold tier is exposed through the GATES API gateway and geospatial portal. Bronze and silver tiers are accessible solely to platform engineers and authorized data stewards respectively.

Zone	Description	Characteristics	Access Scope
Raw (Bronze)	Stores extracted data from the ingestion layer	- Minimal Transformation - Source preserving - Restricted Access - Used for audit and replay - Not intended for broad consumption	Platform Engineers
Validated (Silver)	Stores data that has passed basic checks.	- Required columns exist - Data types are valid - Geospatial fields are valid - Duplicate records are handled - Basic quality rules pass	- Data Stewards - Approved Pipelines
Curated (Gold)	Stores cleaned, modeled, and analytics-ready datasets.	- Stored as Iceberg tables where applicable - Documented in DataHub - Owned by a defined team or steward - Subject to data quality checks - Suitable for dashboards, notebooks, and APIs	Approved Pipeline
Published (Gold)	Stores approved data products.	- Approved for internal or external consumption - Has complete metadata - Has defined access policy - Has lineage and owner information - May be exposed through APIs, dashboards, or GIS Applications	- API Gateway - External Partners

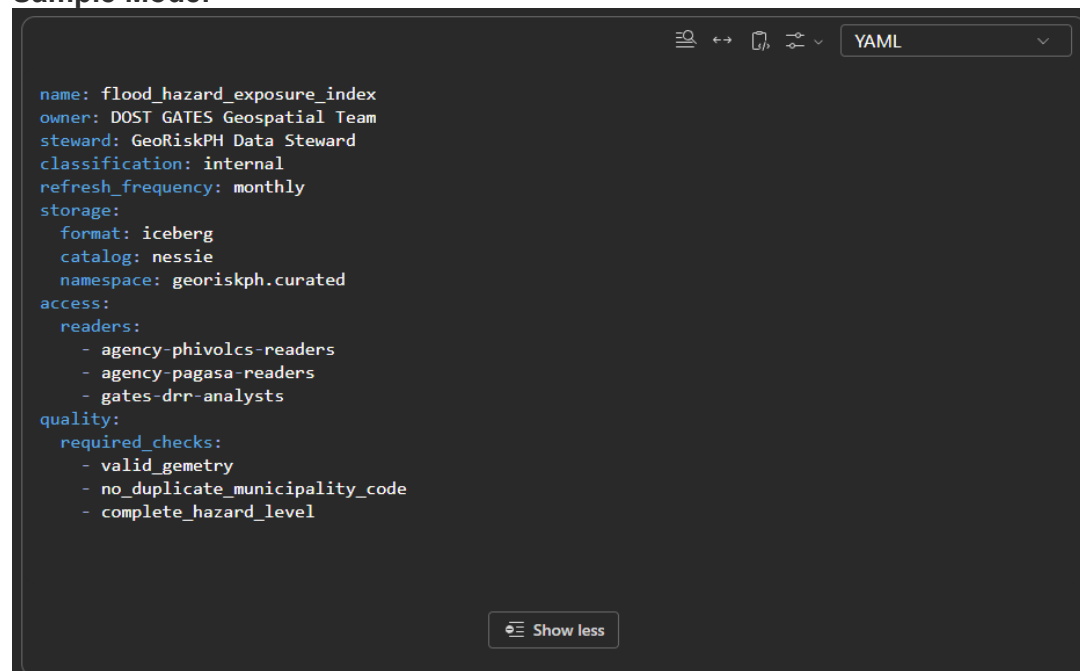
Data Product Model

Metadata of a production grade dataset should have defined governance rules. Each dataset must be documented, owned, and reusable.

Data Product Metadata:

- Name
- Description
- Owner
- Steward
- Source system
- Business meaning
- Schema
- Classification
- Access Policy
- Refresh Frequency
- Quality expectations
- Lineage
- Contract or support channel

Sample Model



```
name: flood_hazard_exposure_index
owner: DOST GATES Geospatial Team
steward: GeoRiskPH Data Steward
classification: internal
refresh_frequency: monthly
storage:
  format: iceberg
  catalog: nessie
  namespace: georiskph.curated
access:
  readers:
    - agency-phivolcs-readers
    - agency-pagasa-readers
    - gates-drr-analysts
quality:
  required_checks:
    - valid_gemetry
    - no_duplicate_municipality_code
    - complete_hazard_level
```

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4. Governance Framework

GATES P2 framework covers dataset registration, access control, data classification, and lifecycle management across the medallion architecture.

Dataset Registration and Onboarding Workflow

1. Agency or project team submits a dataset registration request.
2. Designated data steward reviews dataset ownership, classification (public / restricted / confidential), and intended use.
3. Platform team provisions storage, namespace, and access policy.
4. Ingestion pipeline lands data into the bronze (raw) zone.
5. Automated validation jobs check schema conformance, null rates, and format compliance.
6. Curated tables are created in the silver zone upon successful validation.
7. Metadata and lineage are registered.
8. Data products are reviewed by the data steward and project lead.
9. Dataset is promoted to the gold (published) zone.
10. Downstream consumers request access through access portal.

Namespace and Tenancy Model

Cluster Group

- Staging
- Production

Major Group

- Platform
- AI
- Agencies

Minor Group

- Agency acronyms/codes in lower case
- Project acronyms/codes in lower case

5. Data Quality Scoring

Scoring Consideration (Platform Perspective)

Platform Team assumes that all data received is already cleaned, validated, and correct. For a dataset to be considered ready for visualization:

At the Backend/API level:

- Data structure matches the expected schema
- Data fields are sufficient to display
- Data is in a platform-consumable format (e.g., GeoJSON for GIS, structured JSON for dashboards)
- API endpoints are stable and returning expected responses

At the Frontend level:

- Required display fields such as coordinates, labels, and timestamps are present
- Data format is compatible with the target platform component (map layer, chart, table)

Post Data Quality Assessment (AI Perspective)

- Data provides complete and objective answers at evaluation
- Consistency across dataset types
- Availability of schema/data dictionary/directory/metadata
- Completeness report/EDA

7. Use Case Details

Designing an AI Use-Case

I. AI Workflow

- data source (flat files, structured database, images, combination)
- objective (questions/problems that need answers)
- success indicators
- complexity (which problems can be addressed by an AI-driven approach)
- LLM capabilities/evaluation results
- Hardware/compute requirements

II. Agentic vs Function/Tool Calling

- autonomy requirement (complex combination of tools, API calls, and agent invocation that require autonomous decision making)
- straightforward workflows with fixed/adjustable input parameters

III. Responsible AI Usage

- Is it fair?
- Are the decisions explainable?
- Document/data provenance
- Who is accountable if it fails?
- What decisions does it influence?
- Data access and security, data protection compliance
- Auditability, output monitoring and traceability
- Human oversight, users can intervene
- Transparency, users must know that AI is being used
- Risk-tiered approach (low risk: minimal controls; medium risk: require human review; high risk: require formal risk assessment, bias testing, human approval, audit logs, appeals process)

9. GATES Organizational Function (BuilGATES)

The GATES P2 team is hosted at DOST-AST. The following table reflects the current team structure, reporting lines, and resource allocation.

Role Assignments

- All technical roles are placed in their technical corresponding teams (Lakehouse, AI, Platform).
- No technical roles are embedded within the Management & Policy Team.

Cross-Functional Processes

- Certain processes require collaboration across teams, for instance:
 - Procurement - both management and technical roles participate; management handles administrative and compliance requirements, while technical teams inform technical specifications and requirements
 - Project Documentation & Input - shared responsibility across teams
 - Decision-Making - key decisions involve alignment between management and technical leads, ensuring both operational and technical perspectives are considered

Stakeholder Request Flow

- Management acts as the primary filter, ensuring requests are properly assessed and discussed before reaching technical teams
- Exception: in engagements where no management personnel are present, requests may be relayed directly to technical teams (but this is rare and non-standard).